G1 Box Grasp: conveyor catch and two-hand grasp

Work report for the LuckyEngine G1 controller `g1_grasp.cs` (namespace `G1Pipe`, class `G1BoxGrasp`) · covers 16–30 October 2026 · repo `~/projects/robot_code/G1BoxGrasp`, engine project `~/Documents/LuckyEngine/Projects/g1_sort`

Summary

The controller started this period as a pick-and-place pipeline for a box *resting* on the conveyor. The conveyor was turned into a gravity slide and a **catch** was built: the left hand predicts the sliding box, waits in its path and gives with it (sections 5–8). The grip then took most of the work. A high-priority hand friction class and `noslip_iterations=20` stopped the box creeping out of the hands, and the carry became walker-only (back off, turn in place, walk straight), so by 25 October the whole pipeline worked for the 0.3 m, 1 kg box (commit `0ec78d3`). Making it work for other sizes and weights (sections 13–17) needed a palm-led squeeze with capped palm and finger forces, so the 5 N·m wrists are never twisted open. It also needed a **waist lock**: the walker's waist yaw was swinging the torso ±25° at about 2 Hz in every turn and flinging the box. On top of those came slower carry speeds, a box raised clear of the table, a centring push that puts any caught box on the body's centre line, and elbows eased outward. On 27 October boxes of 0.2, 0.3 (1 kg and 2 kg) and 0.4 m were caught, carried and placed. The latest 0.1 m run reached the table but slipped out as Place began. The remaining weak spots are slow placing (the turns at the table take two tries), a heavy (2 kg) box sinking 4.4 cm of the 5 cm allowed, and the narrow box's grip. Section 18 is the plan for the next size and weight tests.

From 28 October (sections 20–28) the work moved to heavier and taller boxes and then to collecting imitation-learning data. Heavy boxes walk the body backwards, so the box is now held in pelvis coordinates from the lift, a calm station hold steps the body back onto its spot, the waist stays locked through Place, tall boxes are carried clear of the head, and placing simply lets go over the table (commit `1515f8d` “Good enough”). An ACT data pipeline was planned and written. On 30 October recording started working: LuckyEngine 2026.1.2 has no Custom Schema recorder mode, so the pipeline reads *MujocoFull* sessions and derives the IMU from the pelvis pose; an episode runner records each run, writes a sidecar of what the recorder can't see, and resets the scene so every episode starts exactly as Play does (it took four attempts, the key being a fresh walker on the same frame of the run). Box attributes are randomised with a pool of six boxes dealt like cards, plus a 0–10 cm start jitter. Recording exposed a DragIn runaway and a waist lock that snapped the torso straight; both were fixed. Several grasp tweaks were tried and reverted, and the grasp code is back at the `82bc6f9` baseline. The final 12-episode recording succeeded 8 times with no squeeze failures; 0.2 m boxes slipping at the start of the carry turn are the main remaining problem.

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1. Project, tools and workflow

Robot and engine. Unitree G1 with three-finger hands (`ContentVault/Robots/UnitreeG1ThreeFingers/g1_three_fingers.xml`) in LuckyEngine (Hazel editor with MuJoCo physics). The legs and waist are run by learned policies in slots: walker = 1, rotator = 2. The arms use a motion-graph LimbIK whose end effectors are `left_grasp_site` and `right_grasp_site`. The IK owns the shoulders and the elbow. The script owns the wrists (a closed-loop wrist servo) and the thumb and finger bases.

Files. The controller is a single file, `g1_grasp.cs` (about 3,260 lines, 83 inspector fields). It's developed in the repo and copied into the engine project's `Assets/Scripts/Client/Source/`. The only difference between the two copies is `ArmReach` (0.50 in the repo, 0.65 in the engine). The phase diagram is `g1_grasp_flow.svg`. Logs are in `/tmp/LuckyEngine/Logs/APP.log`, and every controller line starts with `[G1]`.

Workflow used throughout. Each change was edited in the repo, formatted with `csharpier`, compile-checked with `dotnet build G1BoxGrasp.csproj` against the engine's `Hazel-ScriptCore.dll` (every build ended with 0 errors), and synced to the engine copy. The user then ran the scene, and the next step was decided from the log. Analysis-only requests were answered without code changes. Commits were made only when asked.

2. The pipeline and its phase loop

A phase machine drives everything: `Advance()` is the only place the phase changes, every phase has a progress watchdog, and `Fail(why)` leads to `Recover`. Figure 1 is the black-and-white loop diagram made in this period, from the generator script `flow.py`, checked by rendering with headless Chrome. It shows the resting-box pipeline. The catch path added later enters it at `PreGrasp` (section 8, Figure 3).

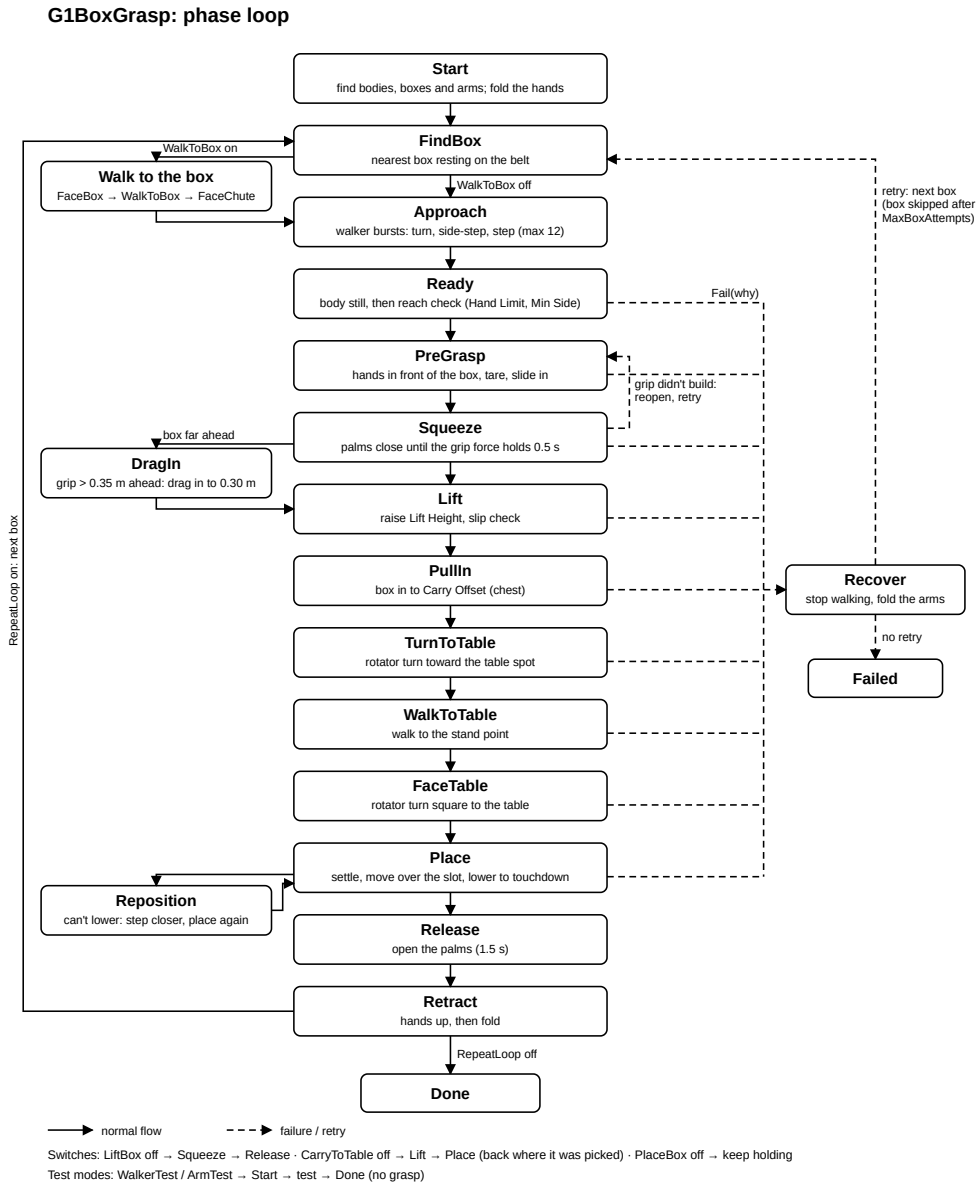


Figure 1. Phase loop of the resting-box pipeline (`g1_grasp_flow.svg`). Main chain: Start → FindBox → Approach → Ready → PreGrasp → Squeeze → Lift → PullIn → TurnToTable → WalkToTable → FaceTable → Place → Release → Retract. Optional walk to the box, DragIn and Reposition on the left. The dashed Fail(why) bus leads to Recover, then a retry from FindBox or Failed. A failed squeeze retries through PreGrasp. The footer lists the phase switches and test modes.

The grasp itself works like this. `Reachable()` plans the grip and checks reach: every hand target within *HandLimit*% of the arm, measured from the standing pose, and each grip point at least *MinSide* off the body's centre line. Both hands glide to **pre-grip** points 10 cm in front of the box, 3 cm wider than the grip. After a tare they **slide in** along the side faces, with *ApproachGap* of air between each palm and its face. **Squeeze** is admittance control on palm force: the gap target goes negative down to $-$*MaxSqueezeDepth* until the palm-only force reaches the required squeeze, factor × m·g / (2μ). The index and middle fingers curl while the palm is on its face. A grip that stalls at full depth is reopened and retried (*MaxSqueezeTries*).

3. Inspector exposure and tuning (resting-box pipeline)

Most of the requests in the first part asked for a constant to be put in the inspector so it could be tuned without code. Everything added is listed in Appendix A. The decisions that mattered:

- **HandLimit vs IKTargetLimit.** One limit was split into two. *HandLimit* (93) is the planning limit, measured from the standing pose. *IKTargetLimit* (95) is the per-frame cap, measured from the live shoulder: any IK target beyond it is pulled in toward the shoulder. For a short time the cap was tied to HandLimit. At HandLimit 110 that let targets reach 110% of the arm, and the result was the singularity described in section 4. The user's point that ">100% is sometimes not unreachable" was handled by keeping the plan limit tunable while the cap stays under 100.
- **GripForceLimit** (100%). The fraction of the required squeeze each hand must hold before the grip counts, so a squeeze that falls slightly short no longer stalls.
- **GripComfort** (85%) replaced `k_Comfort`. The grip is raised up the box until each grasp site is within this share of the arm.
- **Thumb angles**, all clamped to the joint limits from the MJCF:
  - *ThumbOpenDeg* (41.25, thumb_1 out);
  - *ThumbGripDeg* (0, thumb_1 in);
  - *ThumbTipDeg* (0, thumb_2).

  The thumb-related ranges are thumb_1 −0.724…1.047 rad and thumb_2 0…1.745 rad. Thumb_0 (±1.047) isn't driven by the script.
- **Tolerances group.** Fourteen hidden constants became fields (ReachProgress, the four Placed* checks, TopMargin, the three Align* values, MinSide, MaxGripUp, StillSpeed/StillTime, TouchForce, Slip/Drop tolerance and LostDistance), plus ReachTolerance moved here.
- **MaxSqueezeDepth** (0.03 by default; the scene used 0.05). How far past its face a palm target may go, and so how much arm tracking error a squeeze can make up.
- **Palm height.** The answer to "which variable decides how far down the box sides the palm goes": *BeltClearance* (Box group). The grip height is `BeltClearance + PalmBelow + (1+Side)·k_ThumbStagger + GripLift`, so the left hand sits 3.6 cm higher than the right so the straight thumbs pass each other.

That stage was committed as `a571ea4` "good enough pipeline". The commit also includes the redrawn phase diagram.

4. Log analyses of the resting-box pipeline

Run / question	What the log showed	Explanation / outcome
Six runs on 16 Oct	2 successes. Short side-step bursts stalled for a box far off-centre. The rotator's carry turn made the box slip. A narrow box was blocked by MinSide and looped. One run was a stale build.	Several fixes proposed but not done: escalate stalled bursts, gentler carry turns, stop chasing a pushed box during PreGrasp, wrap the thumbs after the palms are on.
HandLimit > 100 (110)	The slowdown and twisting started on the frame the per-frame IK target crossed 100% (right hand 1.04 m/s, sim 0.13× real time). Then the hands pushed the box, which moved the goals, which pushed the box again.	A target beyond full stretch is an IK singularity. Fixed by the separate *IKTargetLimit* cap at 95%.
17:42, cap at 95%	No slowdown. Failed "right hand short": the hand was 0.6 cm over the box top against PlacedTop 0.01, with ReachTolerance 0.05 borderline.	The tolerances were too strict, which led to exposing all of them (section 3).
15:54, box halved (0.152 wide)	Rejected by MinSide 0.11: a grip point sat 0.10 m from the centre line. After MaxBoxAttempts = 5 the box was skipped and FindBox stalled.	The arm sweeps showed IK accuracy falls off within about 0.12 m of the centre line, so narrow boxes were blocked by design.

Run / question	What the log showed	Explanation / outcome
16:00, squeeze not working	ThumbGripDeg 15 drove the thumb tips into the front face. They pinned the hands with friction, pushed the box 3.3 cm away and tipped it. The palms stayed 2.5–3 cm off. The right arm's IK landed 1.9 cm outward on the narrow box, against 0.5 cm inward on the wide one.	Keep the thumbs straight (0) for the squeeze.
Narrow box always leaves a gap	The same gap is commanded for both widths, but the arm's tracking error near the centre line uses up the squeeze depth.	*MaxSqueezeDepth* exposed.
16:41, 0.15 box, depth 0.05	Approach left the box 4.4 cm off-centre, inside AlignSide 0.06. The left grip point was 0.07 m from the centre line. During the slide-in the left target moved 3 cm inward but the hand didn't move: lag 0.9 → 4.6 cm, then 7 cm in the squeeze. The left palm stayed 6.6 → 4.2 cm off its face. The right hand gripped (fingers 34 N) and pushed the box, which tipped on the retry.	More depth only moves the target; the arm doesn't follow that close to the centre line. Suggested: AlignSide about 0.015 and MinSide back to 0.11. The real fix would be finer centring steps and checking MinSide at the final grip points instead of the pre-grip ones.

5. Turning the conveyor into a slide

5.1 What the scene had

Entity	Contents	State
ConveyorBelt	`G1Sort.BeltForce` : speed 0.3, direction +Z, for PoolBox_01..10. Writes box qpos each step.	Did nothing: it looks bodies up by tag, but scene bodies are named `lucky_body_<entityID>` . Even working, a qpos-driven box would pass through a hand.
ConveyorGateController	`G1Sort.ConveyorGate` : weld/unweld a `Trapdoor` body every 1 s.	Did nothing: no Trapdoor entity, and the same tag lookup.
FunnelWall_01/02/03/East	A vertical funnel at x 0.80–0.95, z 0.575–0.725, height 0.88–2.58.	Inside width exactly 0.15 m, the box width, so a box would jam. It sits over the far side of the belt.
SpawnMarker	5 cm cube *with a collider* inside the funnel.	Would catch a falling box.
Belt geometry	x 0.44–1.02 (0.58 wide), z −0.735… 1.011, top at about 0.49, friction 1.7, tilted 1° with the low end at −Z.	The robot faces +X, so its left is −Z. A box sliding −Z travels toward the robot's left, which is also the table side.

5.2 Script belt vs. gravity slide

Two options were compared:

- **Scripted belt:** a speed controller with a force limit, applied through `SetQfrcAppliedByDof` . The force has to be zeroed when the box is lifted, because MuJoCo keeps applied forces until they're overwritten.
- **Gravity slide:** a tilted belt with low friction, and no script at all.

The user chose the slide, and it's the better fit for the catch:

- the box arrives with real momentum;
- a stopped box only leans on the palm with $m \cdot g \cdot (\sin\theta - \mu\cos\theta)$, about 0.1–0.25 N.

Two requirements came out of the comparison:

1. **Lower the friction on both the box and the belt.** MuJoCo uses the larger of the two frictions, so lowering only the belt does nothing.

2. **Add an end stop** at the low end, so a missed box doesn't fall off.

Tilt	μ (box)	downhill accel	0.4 m/s knock slides	Note
1.9°	0.01	0.23 m/s²	0.82 m	First scene. A light touch sends the box off the belt.
5°	0.06	0.27 m/s²	0.14 m	Same speed, much more damping.
8°	0.11	0.30 m/s²	0.07 m	Keep μ at about 70–80% of tanθ; the pyramidal friction cone tends to stick near the limit.

The belt was later run at 2.7–3.1°. The measured accelerations, 0.14 m/s² at 3.1° and 0.07 m/s² at 2.7°, match $g(\sin\theta - \mu\cos\theta)$ with $\mu \approx 0.04$ (the belt's value). The belt also has a slight twist (Rotation Y 0.0099, Z −0.0016), which makes the box drift about 2 cm outward per metre.

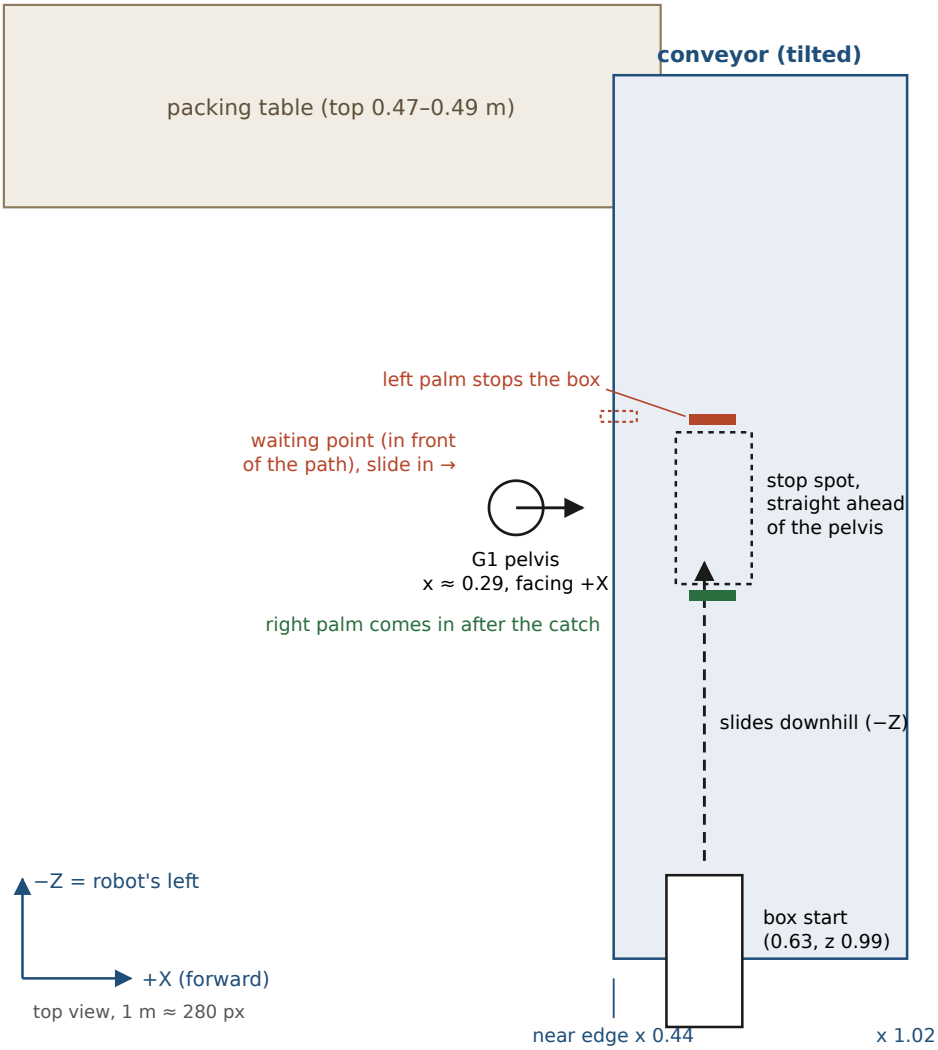


Figure 2. Top view of the working layout (Hazel coordinates, Y up). The box slides along x ≈ 0.63 toward −Z, the robot's left. The left palm closes the path at the box's downhill face. It first waits in front of the path (toward the robot) and then slides forward into it. After the catch, the right hand comes down on the uphill face. The waiting point must stay in front of the belt's near edge (x 0.44).

6. The catch: design iterations

The catch took about a dozen log-driven iterations. Each row below is one change and the run that motivated it.

#	Version	What the run showed	Change made
1	Hold the box with `WeldBody` until the hand is ready, then release	The box started sliding at t=0, ran the whole belt and fell off the end after about 4.3 s. One persistent `WeldBody` call made in `Begin` didn't hold the box. The robot then side-stepped toward a box on the floor.	The user dropped the weld: "assume the box starts at the top and accelerates, track it from the start, predict, put the left hand in between".
2	Track from the first frame and predict; stop spot straight ahead; shift it down the belt if the box would beat the hand	The robot spawned out of reach (pelvis x 0.02, target 151–193% of the arm). The spawn lurch moved the pelvis 0.26 m forward and back and turned it to −34°. The palm came down from above onto the box edge. The box bounced off the stiff palm (0.24 in, −0.50 m/s out) and was pushed outward, x 0.63 → 0.80.	Diagnosis: a stiff arm plus a frictionless belt makes an elastic bounce; the hand came in from above; the robot was too far away.
3	Side approach, give, measured acceleration	—	Waiting point in front of the path, fingertips clear (site k_CatchFront 0.10 + FrontInset in front of the front face; fingertips reach about 8.5 cm past the site), then a slide straight in. A give that runs with the box and brakes over *CatchGive*. Acceleration learned from the box's speed changes.
4	Same	The **right hand flailed**. CatchPose bypassed the fold smoothing, so its IK target was glued to the lurching pelvis, and its wrist servoed to the box frame. The left hand's slide waited on the right hand's moves (the moving-check looked at both arms). It slid in too late, at 103–110% reach, and the fingers flung the box 0.5 m sideways.	The free hand is *parked* (smoothed fold, wrist on the heading). The moving-check looks at the stopper only. The slide starts only if the box is ≥ 0.8 s away.
5	Same	The lurch comes from the spawn: the pelvis starts at 0.79 m and drops to 0.47, then staggers for about 3 s, the same with no catching. The box arrives at 2.7 s. The first-frame shift came from the spawn pose (182%) and could never come back down.	Step 0: wait folded until the body is still, or until the box is so close the hand must go. The spot is fixed in the world once the hand goes. The shift is predicted only once the hand is at its waiting point.
6	Same	"The catching works, but low friction knocks the box away." The shift jumped 15 cm during the slide, turning it into a 25 cm diagonal sweep through the box's lane.	Shift frozen once the slide starts. Slide needs +0.2 s spare (1.0 s in total).
7	5° tilt, μ 0.06	The hand never got settled in time: the first move was planned from the spawn pose as one fixed 2 s move, and every target change restarted a move of at least 0.5 s.	**Continuous tracking**: the stopper's IK command eases toward its target at up to *CatchHandSpeed*, with no restarts.

#	Version	What the run showed	Change made
8	3.1° tilt	"Sudden jerks bounce the box." The give started 0.29 s early, jumped from 0 to 0.45 m/s, and finished before the box arrived, so the box hit a *stopped* palm and bounced (0.49 → −0.35 m/s). The arm was at 87–95% reach. The palm hit above the centre of mass, twisting the box outward.	**Speed-matched give**: palm speed = $v_{box}(1 - \sqrt{(gap/0.05\ m)})$, then braking once the box is on the palm. Stopper palm centred on the box's mid-height. Committed: `81c4fea` "catch works".

7. The catch as it works now

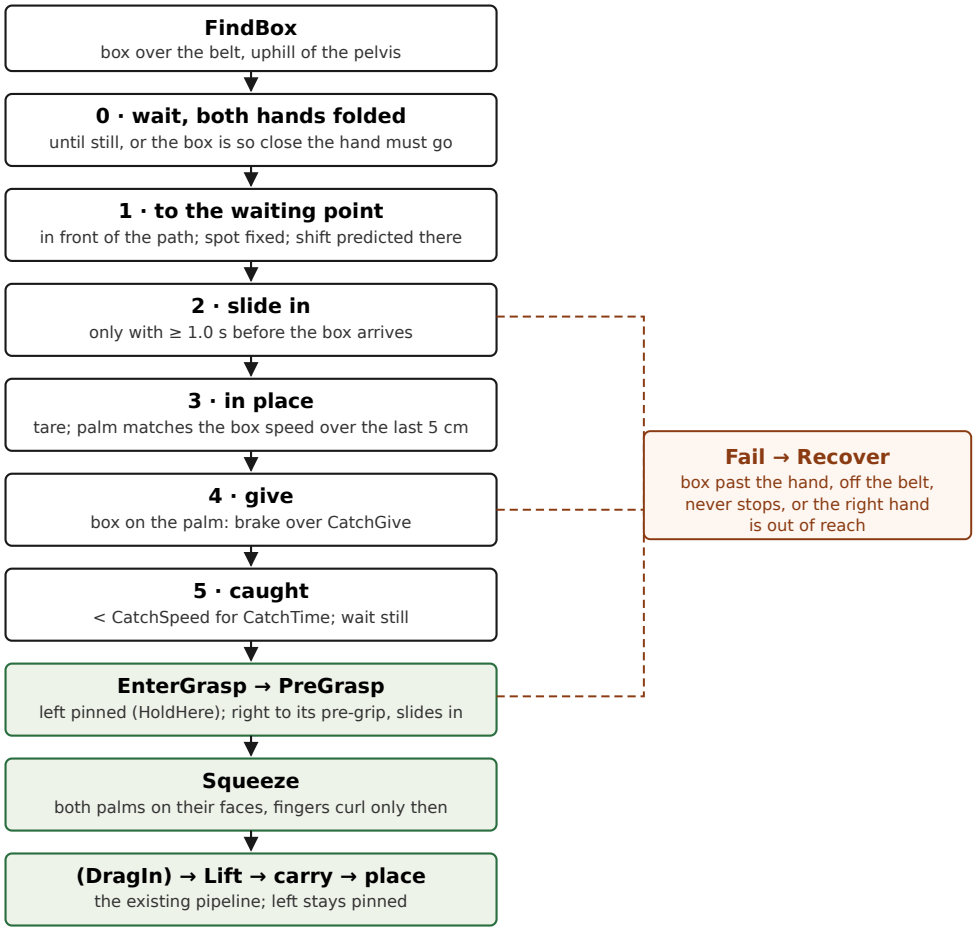


Figure 3. Catch steps (white) and the handoff into the grasp (green). Any step can fail into Recover (dashed).

7.1 How each part works

- **Prediction.**
  - **Path:** the box's path is a line through where it rests on the belt, along the in-plane downhill direction d = normalize($n \cdot n_y$ − up).
  - **Stop spot:** the path point straight ahead of the pelvis.
  - **Hand time:** distance ÷ CatchHandSpeed + about 0.2 s of easing + 0.3 s of arm lag.
  - **Box position at that time:** $u + v \cdot t + \frac{1}{2}a \cdot t^2$, where *a* starts at the frictionless value $g \cdot \sin\theta$ (the earliest the box can arrive) and is then learned from the box's speed changes. It came out at 0.233 m/s² at 1.9°, exactly $\sqrt{(0.33 - 0.01 \cdot g)}$.

- **Stopper motion.** The left hand's IK command tracks its target continuously: it moves at up to *CatchHandSpeed*, eases in near the target, and never restarts a fixed-length move. During the give it follows the moving spot directly.
- **Give.** Over the box's last 5 cm the spot speeds up, so the gap closes at a steadily falling rate and the palm meets the box at about its speed. Once the box is on the palm (force over TouchForce, or within 5 mm) the spot brakes at $v^2/(2 \cdot CatchGive)$. In the good runs, the palm was already moving at 0.19 m/s when a 0.32 m/s box touched it, the impact was 5.4 N, and the box stopped on the spot.
- **Free hand.** The right hand is parked in the smoothed fold, with its wrist on the heading, so it never feeds body sway back into the walker.
- **Start pose.** At spawn the elbows are at 0, which means 90° bent with the forearms pointing *forward*, over the belt when the robot stands close. *StartShoulderDeg* (25, 10, 0) and *StartElbowDeg* (−40) are written straight into the joint positions on the first frame, and the first IK targets are aimed at the fold point.

8. Catch → grasp handoff

Version	What the run showed	Change / decision
v1: holder flag and normal `Reachable()` (commit `8863226`)	The catch was good, but the give carried the box 10.4 cm left and the body drifted 2 cm right. The box ended 12 cm left of straight ahead, so the right grip point sat 0.4 cm from the centre line and MinSide failed. With the box path at x 0.77 the left hand was 9–12 cm short (capped).	The right arm never got a target. The box path has to stay at x ≈ 0.63.
Give-travel planning (start the spot 11 cm up the belt)	Worse. With the box at x 0.715 the waiting point (x 0.49) ended up 5 cm *over* the belt and 3 cm above it. The sagging, fully stretched hand rammed the conveyor (70–237 N, sim 0.2× real time).	**Rolled back** to `8863226`. The patch is saved in the scratchpad (`uncommitted_give_plan.patch`).
v2: pinned holder (commit `9a291ce`)	The catch phase is untouched. The left target is pinned where its IK command was when the box was caught, and from then on only its squeeze, the lift and the carry move it. The right hand gets its own reach check (MinSide only warns) and goes in through PreGrasp. The grasp ran: right hand at 93%, slid in, both "placed". The grip was declared, but the palms were at 0 N: curled fingertips (26–28 N) held the palms off the faces. The box rotated at 0.46–1.1 rad/s and slipped 5.2 cm in the lift.	Rotation diagnosis (section 9). Then: palm-on-face became a live check with finger back-off, and the palms are levelled and kept fully on their faces.
Palm fixes (uncommitted)	The palms now reach the faces, but contact keeps switching between the palm and the flush, straight finger links (about 20 N palm / 0 fingers, then the reverse, every 40 ms). The live palm check flickers, so the grip never confirms. Meanwhile the box is squeezed *out forward* (x 0.667 → 0.885, up to 0.58 m/s) with the palms only 1–4° off parallel.	Effective hand friction ≈ the box's 0.01. Hand friction class added to the MJCF (section 9). Flicker fix proposed (section 12).

Holder mechanics. `HoldHere()` stores the holder's offset from its normal `HandTarget` in grip-frame components (along m_Lat, m_Fwd, up). `HandTarget` adds the offset back for the holder in the current frame, which follows the body while carrying. Two things follow: the holder never jumps when phases change, and it stays pinned through the squeeze, lift, carry and place. While a box is held, PreGrasp and Squeeze don't re-read the box every frame. If they did, the holder's target would follow the box, and a box leaning on the palm would creep away with it.

Right palm height. The right palm goes level with the holder's on a box wide enough for the two straight thumbs to pass (width over about 2 × 0.095 m). On a narrower box it goes 3.6 cm above. Either way it's clamped so the whole palm stays on the face, at least BeltClearance above the box bottom and under the box top.

Palm-driven squeeze, live. "Palm on its face" turns on at TouchForce and off below half of it. The fingers curl only while it's on and ease back at twice the rate when it isn't. The grip counts only if both palms are on at that moment.

9. Friction and the slipping box

> **Why a two-point grip lets the box pivot**
> When only fingertips touch, the box is held at two points and can rotate about the line between them. The only thing resisting that is *torsional* friction, the second friction number: 0.005 m × 27 N ≈ 0.13 N·m per hand. The box's weight (about 9.7 N for the 0.30 box) acts 3–5 cm from that line, a twist of 0.3–0.5 N·m, so the box rotated (logged 0.46–1.1 rad/s). Flat palms on the faces resist this; a torsional value of about 0.05 would too (about 1.3 N·m per hand).

> **Effective hand–box friction is close to the box's, not the hand's**
> With the palms 1–4° off parallel and squeezing 14–28 N each, a contact with μ = 1 can't be pushed out: the box would need a push of about 56 N. Yet the box slid out forward at up to 0.58 m/s. The belt's numbers show MuJoCo combining box and belt with the usual "larger of the two" rule. So the hands must be getting about the box's 0.01. Either the engine gives scene geoms precedence, or the robot's hand geoms have low friction. This wasn't verified directly, because there's no API to read geom friction.

Fix applied (robot model, outside the repo; backup `g1_three_fingers.xml.bak` in the scratchpad). A new default class under `collision`, used by all 18 hand collision geoms: both wrist_yaw/palm geoms plus thumb_0/1/2, index_0/1 and middle_0/1, per hand:

```
<default class="hand">
  <geom priority="2" friction="1.5 0.05 0.0001" condim="4"/>
</default>
```

With a higher `priority`, MuJoCo uses that geom's friction outright instead of combining the two. The hands therefore grip with μ = 1.5 and torsion 0.05 whatever they touch, while the box-on-belt contact keeps its low friction. `condim="4"` makes the torsional value take effect. This change is not yet tested.

10. Engine facts and gotchas learned

Topic	Fact
Scene body names	Scene entities with MuJoCo colliders become bodies named `lucky_body_<entityID>`. `GetBodyID(tag)` fails for them, which is why the old BeltForce and ConveyorGate scripts did nothing.
Frames	MuJoCo-frame poses are converted with `Conjugate(GetAxisHzToMj())·p + GetSceneTranslation()`. Hazel is Y-up; yaw = atan2(−z, x); left of a heading = (z, 0, −x).
Weld API	`WeldBody(uint bodyID, Vector3 positionMj, Quaternion orientationMj, bool persistent = true)` and `UnweldBody`. A single persistent call in `Begin` didn't hold a box. The user's own older scripts re-assert the weld every frame.

Topic	Fact
No velocity or friction setters	Scripts can set qpos, ctrl and applied forces (`SetQfrcAppliedByDof`), but not qvel or geom friction. Bodies can't be added after the model is compiled, so use a pool of pre-placed boxes instead of spawning.
Friction combination	Default: per component, the larger of the two geoms' values; with `priority` , the higher-priority geom's values. The engine uses a pyramidal friction cone (its log warns of a "Velcro effect"). Torsional friction only applies with condim ≥ 4.
Spawn dynamics	The pelvis spawns at 0.793 m with straight legs, drops to about 0.47–0.50 within 0.5 s, lurches about 0.2–0.3 m forward and back, and settles after about 3 s, with or without arm motion.
Elbow zero	Elbow 0 is 90° bent with the forearm level and forward; +90° is straight; −60° is folded up.
IK cap	The capped target is pulled straight in toward the shoulder, so capping is smooth. Jerks at 90%+ reach come from the near-straight arm's sensitivity, not from the cap.
Motion graph	The `Start` trigger stores the current pose and blends to the IK solution over `Duration` ; after that, targets are tracked every frame.
Contact sensing	Hand contact force is read per geom and tared. With straight fingers, the palm and the proximal finger links are flush, and the reading switches between them from frame to frame.

11. Scene layout rules

- **Box path in reach, but in front of the belt edge.** The waiting point is about 15 cm in front of the box's front face and must stay in front of the belt's near edge (x 0.44). So the box centre should be at x ≤ about 0.66; x ≈ 0.60–0.63 works.
- **Pelvis about 0.27 m behind the box's front face**, x ≈ 0.28–0.31 for a box at 0.63. That gave a comfortable 80–83% reach. At a box path of 0.715–0.80 the left hand was capped at 104–115% and stayed 9–15 cm short.
- **Arrival time.** Box arrival should come after the spawn stagger (about 3 s). Lower the tilt slightly or start the box further up the belt.
- **Friction.** Keep the box's sliding μ below tanθ (about 70–80% of it). A far-side rail, 3–5 cm high at x ≈ 1.03, catches knocked boxes.
- **Belt twist.** Zero the belt's Rotation Y and Z (0.0099, −0.0016) so the box slides straight.

12. Commits and state on 19 October

This section records the state at the end of the first period. Sections 13–19 cover what came after.

Commit	Date	Message	Contents
`a571ea4`	18 Oct 19:32	good enough pipeline	The inspector fields from section 3 and the redrawn phase SVG.
`81c4fea`	18 Oct 21:32	catch works	CatchBox phase: prediction, waiting point and slide-in, speed-matched give, parked free hand, continuous tracking.
`8863226`	18 Oct 23:08	Catch works perfectly; the other arm is still not coming	Tucked start pose. Handoff v1, blocked by MinSide.
`9a291ce`	19 Oct 21:20	Catch working perfectly; right arm comes in but the box slips	Pinned holder (HoldHere). Right-hand reach check. Holder kept through the carry.

Uncommitted right now:

- in `g1_grasp.cs` (29 lines added, 21 removed): the live palm-on-face check with finger back-off, and the right palm levelled and kept fully on its face;
- in the robot model: the hand friction class (outside the repo).

Next steps (as planned on 19 October)

1. **Test the hand friction class.** Expect no more squirting, and much less pivoting.
2. **Palm-contact flicker.** While the fingers are nearly straight (curl under about 0.1 rad), count palm plus finger force as "flat on the face". Still require palm-only contact before the fingers curl further.
3. **Robustness: the next topic you raised.** Make the catch and grasp work for different box sizes, not just favourable ones, and still work with zero torsional friction and when the box turns on impact. Candidate pieces:
   - re-read the box's yaw and centre after the catch, and re-align the grip frame and the right hand to it;
   - pick the stop spot and grip heights from the box's size;
   - plan the give's travel without letting the waiting point cross the belt edge;
   - check reach before the hand goes, and fail early with "move the box or robot by X";
   - a squeeze that resists rotation: palms at the same height, through the box's centre of mass, with the fingers kept straight until both palms carry load.
4. **Message fixes.** The "box sits 0.25 m too low" warning really means "too far ahead". A palm that never reaches its spot should report that, instead of "never stopped" after 15 s.
5. **Repeat loop.** A pool of pre-placed boxes, or an end stop and recycle, for *RepeatLoop* with the catch.

13. 19–25 October: a carry that works

Problem	What the logs showed	Fix
Box squirted out of the squeeze	The hands gripped with about the box's own friction (0.01).	The hand friction class from section 9 (`priority=2`, μ 1.5, torsion 0.05, condim 4) worked: the box no longer squirts, and the whole pipeline ran with the box at a favourable position (commit `21de146`).
Slow creep in the hands	Over a long carry the box crept down the palms by millimetres per second, even with both palms on.	`noslip_iterations="20"` in the robot MJCF's `<option>`. MuJoCo's soft contacts let a squeezed box creep; the noslip solver removes that. Commit `12d4f2e` "Slippage problem fixed".
Jerky walk to the table	The rotator's take-over lurched the body, and an IK-led turn dragged the box.	Walker-only carry: **BackOff** straight back from the conveyor, **turn in place** on the walker (the rotator lurches the body when it takes over), a straight **walk**, then **FaceTable**. A failed turn never drops the box; the robot goes on from where it is.
Shoulders and elbows oscillating while turning	The hand targets followed every step of the gait.	The hand targets ride a smoothed carry frame (*CarrySmoothing*, 0.2 s low-pass of the pelvis pose and heading). The wrists hold their angles from grip confirm onward.

With these, the 0.3 m, 1 kg box went through catch, grip, lift, back-off, turn, walk and place reliably, and the user committed it as `0ec78d3`. The next goal was set then: *different box sizes and weights*.

14. The grip as it works now

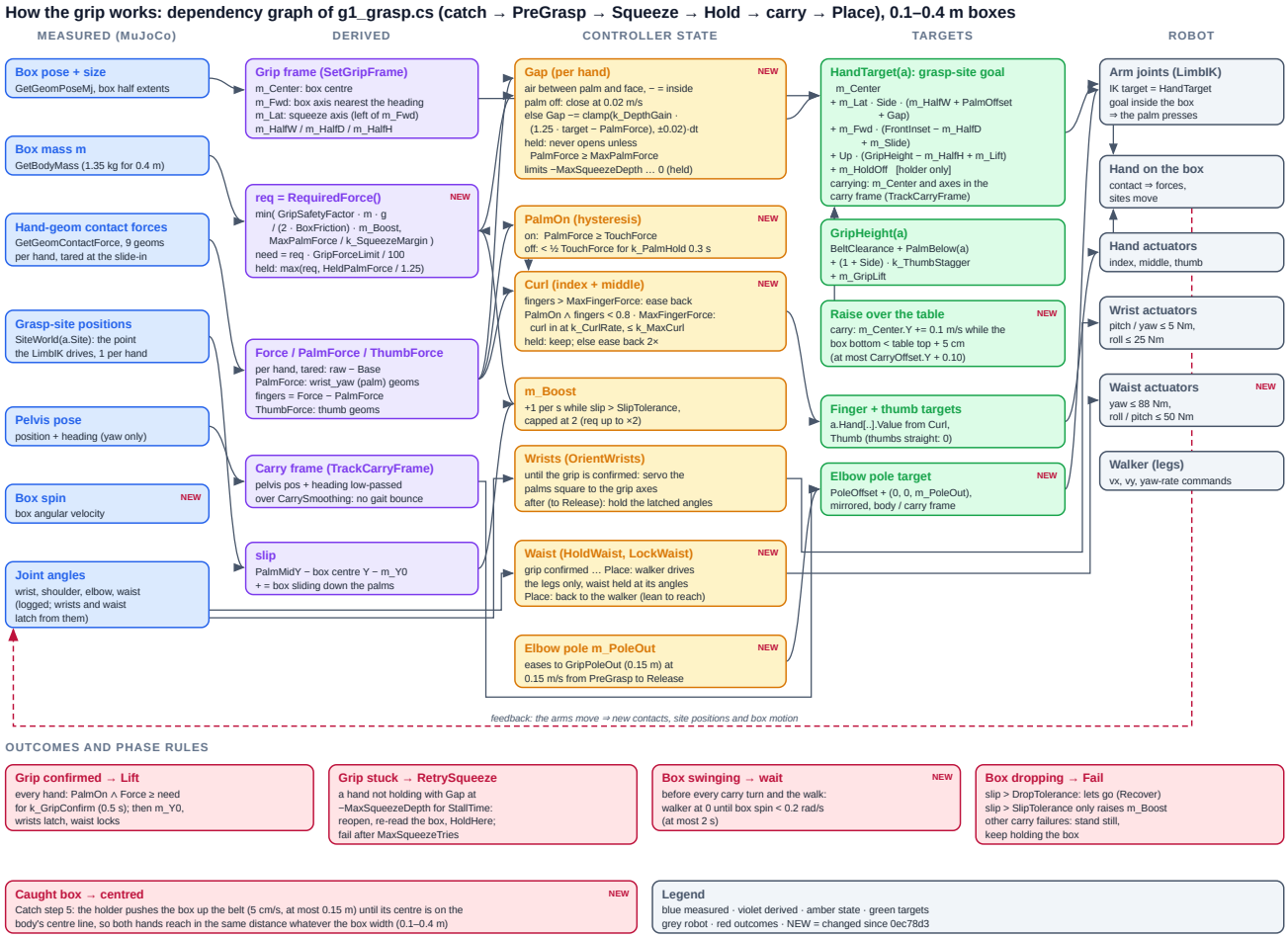


Figure 4. Dependency graph of the grip (`grip_dependency.svg`), from the measured quantities through the controller state to the targets the IK and actuators get. NEW marks what changed since `0ec78d3`.

Every piece below came from a log. The constraint throughout: the wrist actuators stay at the MJCF's 5 N·m (pitch and yaw), because raising them would widen the sim-to-real gap.

- **Palm-led squeeze.** The gap of each hand is an admittance on the *palm's* force only. While held, a palm that comes off closes straight back in, and the hand never opens unless the palm presses past *MaxPalmForce* (40 N). While holding, each palm aims at *HeldPalmForce* (20 N), or more if the box's weight needs it. Palm-on has a 0.3 s hold-over, so carry jostle doesn't make it flicker.
- **Capped fingers.** Index and middle curl in while the palm is on, but never press past *MaxFingerForce* (25 N): over it they ease back, and under 0.8× it they curl on. They press 16–22 cm out from the wrist, twice the palm's lever. Uncapped, they reached 60–100 N and dragged the wrist yaws 50–57° off their commands (right −11° → −68°, left 5° → 53°), which lifted the palms off. With both caps, the wrists stay within about 4° of their latched angles.
- **Why not a full finger wrap.** With *FrontInset* 0.04 the knuckles sit 4 cm behind the box's front face. Curling then presses the first finger segment into the side face instead of hooking the front corner, so a curl strong enough to wrap is exactly what twists the wrists open. The thumbs stay straight (0); every thumb curl tried pushed the box forward.
- **Centring push (restored on 27 October).** After the catch, the holding hand pushes the box up the belt, at 5 cm/s and at most 0.15 m, until its centre is on the body's centre line. Both hands then reach in the same distance whatever the box width. The 0.1 m box needed this:

caught 8 cm left of centre, its right palm had to reach 3 cm across the centre line, and the right upper arm hit the torso 0.20 m short of its target.

- **Elbows out.** From PreGrasp to Release the elbow hint eases 0.15 m further out (*GripPoleOut*, at 0.15 m/s), so the IK closes the hands by turning the shoulders rather than swinging the upper arms into the torso. Switching it out all at once had jolted the box before, hence the easing.

15. The turning oscillation and the waist lock

With the grip holding, the 0.4 m box still slid out in every turn. New logging of the torso and waist, the shoulder and elbow angles and the grasp sites in body coordinates showed where the swing came from:

Turn time	Pelvis yaw rate	Torso yaw rate	Waist yaw	Left shoulder pitch	Left hand fwd of pelvis
0.02 s	+0.11	+1.16 rad/s	+8°	10°	0.26 m
0.26 s	−0.11	−1.56	−12°	17°	0.30
0.74 s	−0.26	−2.67	−8°	13°	0.30
2.42 s	+0.96	+2.75	−27°	34°	0.28
3.14 s	+1.23	+4.91	0°	24°	0.23

- **The torso, not the pelvis, oscillated.** It twisted on the waist yaw joint 10–20× faster than the pelvis turned, ±25° at about 2 Hz. It started at the end of the back-off, before the turn.
- **The arms followed the twist.** The hands swung ±4 cm fore-and-aft in opposite directions, and the shoulders swung in antiphase as the IK pulled the hands back to targets that barely moved (about 1 cm). Pulling the hands back reacted on the torso and kept the oscillation going.
- **Fix: *LockWaist*.** From grip confirm until Place, the walker drives the legs only (`SetDrivenJoints` without the three waist joints). The script holds waist yaw, roll and pitch at their angles at confirm, with up to 88 / 50 / 50 N·m. Afterwards the torso yaw rate matched the pelvis everywhere, the box's spin stayed at 0.1–0.3 rad/s, and the 0.4 m box reached the table for the first time.
- **Side effect:** with the waist locked, a 1.5 rad/s turn command stalls, so the carry turns now use 2.0.

Two more carry rules came out of the same logs. Before every carry turn and before the walk, the walker stays at zero until the box spins under 0.2 rad/s (at most 2 s). A turn's command eases in over *CarryTurnRamp*, eases down over the last 20°, and ramps out over 0.5 s.

16. Carry speeds and placing

- **Speeds.** Going faster with the waist locked (back 0.6, turn 2.5 rad/s, walk 1.0 m/s) backfired. The back-off overshot to 0.50 m against an aim of 0.20, the turns overshot by 15°, and the robot ended 0.62 m from its stand spot. The settings that work: *CarryBackSpeed* 0.4, *CarryTurnCmd* 2.0, *CarryWalkSpeed* 0.7. The stops allow for the coast: the back-off stops early by speed × 0.4 s (but covers at least half its distance), and the walk stops early in proportion to the square of its speed.
- **Table clearance.** The carried box sagged until its bottom was level with the table top (0.495 vs 0.494 m) and hit the edge, bouncing the robot 0.31 m back. Now, during the turn, walk and FaceTable, the carry target rises at 0.1 m/s while the measured box bottom is less than 5 cm over the table top, up to 0.10 m above *CarryOffset*.

- **Reach for placing.** From 0.45 m away the arms ended fully stretched (shoulder pitch −42° to −65°) with the 0.4 m box still 0.11–0.14 m above its spot. The 0.4 m box's wider hands reach less far forward, and a locked waist can't lean. Now the robot stands 0.33 m from the slot (*PlaceStandDistance*), and the waist goes back to the walker for Place so the torso can lean into the reach.
- **Slot choice.** The slot is chosen nearest the point the robot turns at after backing off, not nearest where it stands at the conveyor. Before this, a far slot made the walk curve back into the conveyor.
- **Placing is slow.** FaceTable takes 5.5–9.6 s, because its turns usually need two tries ending 6–14° off. Place then lowers at 3 cm/s for 2–3.5 s. A whole run takes about 38–48 s, and the table end accounts for 8–12 s of it.

17. Box sizes and weights: results

Box mass is read from the logged squeeze: req = 2·m·g / (2·1), so m = req / 9.81. All the boxes are 0.15 m deep and about 0.15–0.17 m high. All runs are 27 October.

Time	Width	Mass	Code	Result	Max sink in hands	Notes
17:55	0.4 m	1.35 kg	`e03fcde`	placed, 1.2 cm above the spot	2.6 cm	Turns took two tries; the walk ended 0.17 m off.
17:58	0.3 m	1.0 kg	`e03fcde`	placed, 1.8 cm above	1.3 cm	Total about 38 s.
17:59	0.2 m	0.67 kg	`e03fcde`	placed, 2.0 cm above	1.0 cm	Caught 8 cm off centre. The left palm was at full squeeze depth with 2.5 N; the fingers carried it.
18:00	0.1 m	—	`e03fcde`	**failed in PreGrasp**	—	The right hand stopped 0.20 m short: its upper arm hit the torso reaching across the centre line. Led to the centring push and elbows out.
18:01	0.3 m	2.0 kg	`e03fcde`	placed, 0.5 cm above	**4.4 cm**	The palms were on throughout (20–70 N). Sinking built in the turn (+1.5 cm) and the Place (+1.7 cm), close to the 5 cm drop limit.
18:33	0.1 m	0.67 kg	centring + elbows out	**slipped out at the start of Place** (6.0 cm)	—	Pushed 0.146 m, which left it 5.7 cm off centre at the 0.15 m cap. The grip confirmed with the left palm at 0 N (left fingers 25 N) and the right palm at 11 N. It was carried to the table and slipped 0.8 s after the waist was handed back.

18. Test plan: more sizes and weights

Goal

Map the envelope of box width, depth, height and mass that the pipeline handles end to end, find where and why each limit fails, and keep the working cases working as the code changes.

18.1 Method

- **One setting, one run.** The simulation is deterministic: the same code and settings give identical runs. Repeats are only worth it with a deliberate change: the box's start 2 cm up or

down the belt, or the robot's start yaw ±3°. Three perturbed runs per cell are enough to show whether a pass is robust.

- **Baseline.** 0.3 m wide, 0.15 m deep, 0.15 m high, 1 kg, with the inspector values of commit `e03fcde` plus the centring push. Every sweep changes one factor from this baseline. Then come the corners.
- **Per-run summary.** A small log summariser (to be written, e.g. `tools/runsummary.py`) prints one line per run from the `[G1]` lines:
  - phase reached and outcome;
  - catch impact force and give;
  - centring push and the offset left;
  - palm and finger force per hand at grip confirm, and time to confirm;
  - the largest box sink and the phase it happened in;
  - the largest wrist deviation from its latched angle;
  - peak box spin in turns;
  - back-off distance, turn tries and final heading error, walk error;
  - release height and distance from the slot;
  - time per phase.
- **Pass criteria.** Placed with the release under 2.5 cm above the spot and within 3 cm of the slot. The box sinks less than 2.5 cm (half the drop limit). No wrist moves more than 15° from its latched angle. The whole run takes under 60 s.

18.2 Sweeps

Sweep	Values	What to watch	Likely limit and the fix ready for it
Width (1 kg)	0.08, 0.10, 0.12, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45 m	Centring push (cap 0.15 m), grip points near the centre line, right-hand reach, palm vs finger share	**Narrow:** the push hits its 0.15 m cap and the palms barely press. Fixes: a higher cap, or move *CatchPalmSide* in by half the box width so less push is needed. **Wide:** arm span and reach at the table, and slots 0.25 m apart can't fit 0.4 m boxes (raise *PlaceSpacing* to the width plus 5 cm).
Mass (0.3 m)	0.25, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0 kg	Sink per phase, palm force against the 40 N cap, wrist deviation, catch impact	req = m·g at factor 2 reaches the cap (40/1.25 = 32 N) at about 3.3 kg; 2 kg already sinks 4.4 cm. Fixes: *GripSafetyFactor* 3; keep the waist locked until the box is lowered; lower more slowly; for heavier boxes, revisit *MaxPalmForce* against the 5 N·m wrist budget.
Depth (0.3 m, 1 kg)	0.10, 0.15, 0.20, 0.25 m	Palm centre against the box's centre of mass (the box pitching), fingertips past the front face	A deep box hangs its centre of mass behind the palms and pitches; a shallow one puts the palms past its back. Fix: set the grip's fore-aft position from the depth (palm centre over the centre of mass).

Sweep	Values	What to watch	Likely limit and the fix ready for it
Height (0.3 m, 1 kg)	0.10, 0.15, 0.20, 0.25 m	Grip height, clearance over the table, the box tipping in the catch	A short box leaves little face under the palm (*BeltClearance*). A tall one tips on impact and needs more raise over the table (cap 0.10 m).
Corners	width 0.1 / 0.4 × mass 0.5 / 2.0 kg	Everything above	Narrow-and-heavy is the hardest case: a small palm patch and a large weight.
Arrival speed	Box start 0, +0.1, +0.2 m further up the belt	Catch impact, give and the box spinning on the palm	A higher impact may knock a heavy box off the stopper hand; *CatchGive* or a longer give.

18.3 Order

1. Write the log summariser and re-run the baseline to record the reference numbers.
2. Fix the two open failures first:
   - the 0.1 m box slipping at the start of Place: hand the waist back only after the box is lowered, or ramp it back;
   - the 2 kg sinking: try *GripSafetyFactor* 3.

   Re-run both.
3. Width sweep, then mass sweep, then depth and height, then the corners. Stop a sweep at its first two consecutive failures and analyse them before going on.
4. After each code change, re-run the regression set: 0.1, 0.2, 0.3, 0.4 m at 1 kg, and 0.3 m at 2 kg.
5. Placing speed: once the sweeps pass, make FaceTable get there in one try (a lower command near the end, or a heading correction during the walk) and speed up the lowering. The goal is a whole run under 40 s.

19. Commits, state and known issues on 27 October

Commit	Date	Message	Contents
`21de146`	19 Oct 22:33	Whole pipeline works with the box at a favourable position; next: robustness	Hand friction class in effect, palm-led squeeze, full carry and place.
`12d4f2e`	25 Oct 13:21	Slippage problem fixed	`noslip_iterations="20"` (robot MJCF) and the carry changes.
`0ec78d3`	25 Oct 13:54	Pipeline works really well; next: different box sizes and weights	Walker-only back-off, turn and walk; smoothed carry frame; wrists held after confirm.
`e03fcde`	27 Oct 17:58	0.4 m box works: waist lock, palm/finger force caps, slower carry	Sections 14–16, plus FreeSlot from the turning point and arm, site and waist logging.
`82bc6f9`	27 Oct 18:42	Pipeline works for 0.1–0.4 m boxes; placing is a bit slow	Centring push to the body's centre line, *GripPoleOut*, this report and the new grip graph.

Known issues.

- Placing is slow: FaceTable needs two turn tries.
- The 2 kg box sinks close to the drop limit.
- The 0.1 m box's grip rests mostly on fingers (the left palm at 0 N), and the latest run slipped as Place began.
- The carry turns overshoot by about 15°.

- The controller is 3,671 lines with 96 inspector fields. The engine copy still differs only in *ArmReach*.

20. 28 October: heavy boxes, tall boxes and “good enough”

The 27 October pipeline handled 0.1–0.4 m boxes of about 1 kg. The next round of tests made the boxes heavier (density 150 → 300, up to 3.3 kg) and taller. Four new problems came out of the logs, and the fixes below were committed as `1515f8d` “Good enough”.

20.1 Heavy boxes walk the body backwards

With a 2–3.3 kg box held 0.3 m in front of the chest, the walker leans back against the load and takes small steps backwards. In the logged 0.3 m and 0.4 m runs the pelvis drifted 10–20 cm back while standing with the box, so the back-off, the turn and the walk all started from the wrong place and the table spot was missed. Three changes:

Change	What it does	Why
A. Lift in pelvis coordinates	At the start of the lift the grip centre and axes are converted into the pelvis frame (`ToBody`). The box rises `k_ClearLift` (3 cm) first, then rises the rest of *LiftHeight* while it is pulled in toward `CarryX()` at `k_CarrySpeed` (0.1 m/s).	A body pushed back by the weight no longer stretches the arms after it: the box follows the body, and a heavy box comes close to the chest early.
B. Station hold	When standing with the box (after the back-off, through the turns, at the table), if the pelvis has drifted more than `k_DriftTol` (10 cm) from its anchor, the walker steps it back, one direction at a time, until within `k_DriftDone` (4 cm).	Puts the body back where the plan expects it.
C. Waist lock through Place	`lockIt = LockWaist && Phase > Squeeze && Phase < Release` (it used to end at Place).	Given back during Place, the walker swung a heavy box loose: waist yaw ±10°, pitch 8–15°.

20.2 The station hold must stay calm

The first station hold ran everywhere, including the lift and pull-in, and corrected forward/back, sideways and yaw at once. Two failures followed:

- **A to-and-fro limit cycle during the lift.** The correction steps rocked the body ±6 cm with a period of about 1.2 s, and flung a 3.3 kg tall box. The user's words: “the robot does a to and fro motion destabilizing everything, it wasn't happening before your weight updates”.
- **Stumbles in the back-off and turns.** Combined commands (vx + vy + yaw, or sideways while still coasting) dropped the pelvis to 0.50 m and swung the heading 19°.

The final rules: no station hold in Lift or PullIn; a correction starts only when the body has been still for *StillTime* and is not turning; it moves in **one direction only** (forward/back if that error exceeds 4 cm, else sideways); it stops as soon as the remaining error is under 4 cm, when overshot, or when a turn starts. The anchor is the grip spot until the back-off ends, then the back-off spot, then the stand spot at the table.

20.3 Tall boxes: the right wrist and the head

- **Right wrist wound to its limit.** On a tall box the parked right hand's wrist servo kept steering toward an orientation it could not reach and wound the wrist yaw to 73–93°, so the hand pointed down. The fix of the day: the wrists of hands that are not gripping, or are parked through the catch, ease back to neutral instead of steering. *(Reverted on 30 October, section 25.4.)*
- **The box hit the head.** A tall box pulled in to *CarryOffset* reached the head mesh (0.33–0.53 m above the pelvis, front at +0.078 m) and the contact destabilised the body. New field

HeadClearance (0.25 m): a box whose top would reach the head is carried with its near face that far in front of the head:

```
CarryX() = CarryOffset.Y + k_MaxRaise + halfH > k_HeadBottom (0.30 m)
           ? max(CarryOffset.X, k_HeadFront (0.08 m) + HeadClearance + halfD)
           : CarryOffset.X
```

CarryOffset's default moved from (0.30, 0.10, 0) to (0.24, 0.10, 0). The scene keeps its saved (0.3, 0.1, 0) until changed in the inspector.

20.4 The parked right arm and the passing box

While the left hand caught, the folded right hand sat where a box sliding along the belt could clip it, and in one 0.2 m tall-box run it pushed the box further out. Two changes were made: `ParkPose` pulled the parked hand back so its straight fingertips (`k_FingerTip` 0.09 m) stayed `k_ParkClear` (4 cm) behind the passing box's near face; and, after an elbow pull to avoid the box made the whole body oscillate, the parked hand was limited to `k_ParkSpeed` (5 cm/s). *(Both reverted on 30 October, section 25.4.)*

20.5 Placing, simplified

At the table the end effectors got stuck between the box and the table while the controller searched for a free slot and stepped closer, and the arms reached their 95 % IK cap. The user asked to drop the slot search for now ("if the box is above the table, just release it there") since ACT will learn placing from the demonstrations anyway:

- `FreeSlot`, `StepCloser`, `StepReposition`, `Phase.Reposition` and the fields *PlaceSpacing*, *MaxRepositions*, *RepositionStep*, *RepositionTolerance*, *MinStandDistance* were removed.
- The walk goes to `TableSpot(from)`: the nearest point on the line *PlaceEdgeInset* in from the table's near edge, clamped to its ends.
- `StepPlace`: settle, convert to world coordinates; if the box centre is already over the table (`OverTable`, 5 cm inset) go straight on, else move horizontally over the spot; then lower at 3 cm/s until 1 cm above the table or 2 cm below the start, and release.
- Release opens the palms at 0.3 m/s to a 10 cm gap and waits 0.5 s (was 0.05 m/s, 5 cm, 1.5 s).

20.6 State on 28 October

The 0.4 m box at density 300 was caught, carried and placed. The user judged it "good enough" to start collecting ACT data, and it was committed as `1515f8d` together with the data pipeline below.

21. The ACT data pipeline: plan, tooling and explainer

The scripted controller solves the task from privileged state (box pose from MuJoCo, tared contact forces), so it can serve as the demonstrator for an imitation-learning policy (ACT, Action Chunking with Transformers) that sees only cameras and proprioception.

File	Contents
plans/G1_ACT_data_pipeline_plan.html	The plan: what ACT controls (31 upper-body actuators + waist lock + walker command; the RL walker keeps the legs), what the engine records and what it cannot (hence a script "sidecar"), the episode loop, randomisation and recovery data, training settings, closed-loop evaluation over gRPC, risks and milestones.

File	Contents
data_pipeline/	common.py (layouts, readers), qa.py (integrity checks), le2act.py (session + sidecar → LeRobot 3.0 dataset), train_act.sh (smoke / pilot / full), eval_act.py (closed-loop through the luckyrobots Session), make_fake_session.py and tests/selftest.sh (the whole chain without the engine), README.html (code guide). Python env in data_pipeline/.venv (lerobot 0.6.1, luckyrobots 0.5.0, torch 2.11 + CUDA on the GTX 1650).
docs/ACT_explained.html	An explainer of ACT read from the ACT, Mobile ALOHA, HumanPlus, ALOHA Unleashed, Bidirectional Decoding and Real-Time Chunking papers and LeRobot's ACT code: action chunking, temporal ensembling, the CVAE, the network layer by layer, training settings, results and ablations, follow-ups, weaknesses and what it means for the G1 data. Not committed.

The vectors. observation.state = 37: the 31 upper-body joint angles (14 arm, 14 hand, 3 waist) and 6 IMU values. action = 35: the 31 upper-body actuator commands, then waist_lock and the walker command cmd_vx, cmd_vy, cmd_yaw. The phase label (catch, grasp, lift, carry, place, retract) is stored but is not a policy input.

Self-test. tests/selftest.sh <g1_three_fingers.xml> writes a fake 3-episode session, runs QA, the viewer, the converter, a load-back check that actions line up with the commands and video frames are in sync, 20 ACT training steps and an evaluation against a stand-in luckyrobots Session. It passed on 28 October and again after every change on 30 October. A 20-step ACT run on the GPU took about 2 s and 1.25 GB.

22. Recording in LuckyEngine 2026.1.2

The plan and the first scripts were written from the online LuckyEngine docs, which describe a *Custom Schema* recorder mode. The installed editor (2026.1.2) has only three data modes: **RobotArm**, **MujocoFull** and **Drone**. The pipeline was moved to MujocoFull and `make_schema.py` and `schema/` were removed.

MujocoFull column	Contents	Used as
`observation.state` (57)	The whole qpos, named in `meta/info.json`: the pelvis free joint (`floating_base_joint_x` … `_qz`), all 43 joints, then each free box (`<Tag>_x` … `_qz`; 7 more per box).	`q.<joint>`; `pose.pelvis` and `pose.box.<Tag>` (privileged, analysis only); the pelvis IMU.
`action` (43)	Every actuator's ctrl by joint name. The walker drives legs and waist through its own channel, so those read 0; the script's arm, hand and (locked) waist commands are exact.	`u.<joint>`: the first 31 actions.
`frame_flags`	Bits `new_policy_step`, `post_reset`.	QA: a reset may appear only on frame 0.

- **No IMU in MujocoFull.** The six IMU values are derived from the recorded pelvis orientation: gravity in the pelvis frame $R^{T}(0, 0, -1)$ and the body-frame angular velocity from consecutive quaternions. A real G1 pelvis IMU gives both.
- **Alignment.** The recorded ctrl in row k+1 equals what the script wrote at tick k: mean error 3.5×10^{-8} rad on the probe actuator (left wrist roll), against 8×10^{-4} at shifts 0 and 2. The converter pairs state k with ctrl row k+1.
- **Files.** Sessions go to `<project>/DataSessions/session_<date>_<time>/` (one parquet and one 256×256 H.264 video per episode, about 1 MB + 1.8 MB for 34 s). `StopRecording` appends an empty (length 0) episode row to `meta/episodes`, which the reader skips.
- **Head camera.** In the first recording `head_camera` looked straight at the wall behind the conveyor: the box showed only at the frame's edge and the hands only in the corners. The user tilted it down; from then on the box, both hands and the table are in view through every phase (the robot's chest fills about the bottom fifth).

Viewer: **`inspect_session.py`.** One self-contained HTML page per session (written to `~/Documents/LuckyEngine/Inspect/`): per-episode checks marked pass, warning or fail (missing values, frame gaps, tick spacing, resets inside an episode, commands outside the joint ranges, one video frame per row, frozen and black frames, sidecar length, action alignment and how well it is pinned down, waist lock values, phases present and in order, expert outcome), the phase timeline, each camera's frame at every phase start, and hover charts of every joint's angle against its command, the walker command, waist lock, IMU, and the box and pelvis paths. Its frozen-frame threshold was tightened to 0.02 per pixel after it flagged the robot standing still: a live render never changed by less than 0.05.

23. The episode runner and the reset between episodes

`g1_grasp.episodes.cs` is a partial-class file next to `g1_grasp.cs`; the controller itself got only a few hooks (`partial`, `RecordTick` at the end of `OnUpdate`, `StartEpisodeState` on Start frames, the first failure reason, and `Repeat = RepeatLoop && !RecordEpisodes` so a run ends instead of looking for the next box).

- **Per episode.** `Observer.BeginEpisode` once the robot is set up; each tick a sidecar row (`frame, sim_time, task_index, waist_lock, cmd_vx, cmd_vy, cmd_yaw, probe_ctrl`) and `Observer.SetCurrentTaskIndex`; Done ends it as a success, Failed or *EpisodeTimeout* (80 s) as a failure; a JSON header with the seed, box tag, size and mass, start pose, start offset, release error and seconds.
- **Between episodes.** Every private field of the controller is reset by copying it from a fresh instance (reflection; the inspector's public fields and the recorder state are kept), and the next frame is a Start frame again.

Making every episode start like Play took four attempts:

Attempt	What happened
`ResetToInitialState` only	Episode 1 started with the robot facing another way and the box by its feet: keyframe 0 is the compiled model's default, not the scene as Play sets it up.
Reset + restore the qpos saved after the first `Begin`	Box and robot back in place, but episodes 1 and 2 started from the same qpos and still ended differently (a slip in the turn; a grip out of reach). The legs already differed by 0.13 rad on the first frame: the walker policy kept its memory of the previous episode.
Hold the box with a weld while the robot stands still 2 s, then release	Rejected by the user: it changed the catch and broke episode 0 (“box left the grasp: 0.25 m”).
Restore on the Start frame, walker off and on	On the frame Play first sets the robot up, episode 0 saves the full qpos (`GetFullQpos`). On the same frame of every later episode: rotator and walker off, `ResetToInitialState` (velocities, contacts), `SetFullQpos`, `ReseedPolicyPDTargets`, walker on, then `Begin` runs as on Play. Episodes after a reset became identical to the tick; 3/3 succeeded. Committed as `9de7a09` “Data collection works”.

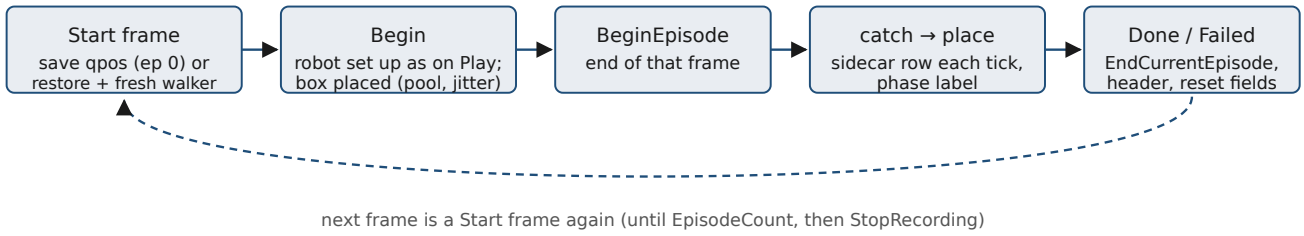


Figure 5. One recorded episode. Every episode, the first too, starts from the same state on the same frame of the run, so starts are exactly repeatable.

24. Box pool randomisation

LuckyEngine 2026.1.2 has no script call to change a body's size, mass or friction at run time (only positions, ctrl and forces). Box attributes are therefore varied with a **pool**: six boxes duplicated from `PoolBox_01` in the editor, one of which is put on the belt each episode.

Tag	Scale (x, y, z)	Density	Mass
Box_W40_L / Box_W40_H	0.15, 0.15, 0.4	100 / 220	0.90 / 1.98 kg
Box_W30_L / Box_W30_H	0.15, 0.15, 0.3	100 / 220	0.67 / 1.48 kg
Box_W20_L / Box_W20_H	0.15, 0.15, 0.2	100 / 220	0.45 / 0.99 kg

Scale z is the box's length along the belt (the squeeze axis). In the last recording the scene had Box_W40_L at 0.2 m, so the pool held three 0.2 m boxes, two 0.3 m boxes and one 0.4 m box.

Getting the placement right took four tries:

Version	Result
Swap the chosen box with the belt box (positions from the Play qpos)	All six duplicates were still stacked on the belt in the editor; on Play they started inside each other and blew apart. The box the robot tried to catch moved at 9.7 m/s and fell off the belt within 0.3 s, every episode.
Belt pose from `GetDefaultFreeJointPosition`	The call does not return the editor pose. The chosen box was placed inside something, shot away at ever-increasing speed and hit the robot on its way out.
Belt pose from the Play qpos, “exactly one box on the belt” check	Worked, but depended on the editor layout: a belt box moved to the belt's top edge was not counted as on the belt and recording was turned off.
BoxStart* and *ParkSpot	The episode's box goes to *BoxStart* (default 0.616, 0.734, 0.978: the original belt start); every other box to its own spot in a row from *ParkSpot* (2 m behind the robot, 0.6 m apart); all upright with the editor-to-MuJoCo rotation (0.707, 0.707, 0, 0). Episode 0 goes through the same reset, so the editor layout no longer matters as long as no two boxes overlap.

Dealing and jitter (commit `0700197`). Random picks with replacement repeated boxes (one session drew `W40_L` three times and never drew the 0.2 m boxes), and every repeat was identical to the tick. The pool is now dealt like cards: each round of six episodes uses every box once, in an order shuffled from the seed. Each episode's start also moves 0–*StartJitter.X* (10 cm) **down** the belt and ±*StartJitter.Y* (2 cm) across it, from its own seed (`BaseSeed + episode`), logged as `start_offset`. The jitter never moves the box up the belt: at the default start the box's centre of mass is only about 3 cm from the belt's top end (a 0.4 m box already overhangs it by about 17 cm), and at the start used in some plain runs (0.623, 0.720, 1.004) it was 7 mm from it, which is why that box was already moving at 1.16 m/s on the first frame.

25. Controller changes found while recording

Recording many episodes of every box size exposed controller problems the single test runs had hidden. Some changes stayed; several were reverted.

25.1 DragIn ran away (kept fix)

When a box is gripped more than 0.30 m ahead, DragIn pulls the grip toward the body until it is 0.30 m ahead of the pelvis, measured every frame. Pulling the hands in made the walker step back, which kept the grip 0.37 m ahead, so the pull never ended: the robot reversed 1.8 m in 7 s at up to 0.55 m/s with the walk command at zero. The drag distance is now measured once when the drag starts (`m_DragLeft`) and the lift begins after that distance, whatever the body does.

25.2 The waist lock snapped the torso (kept fix)

When the grip was confirmed and the waist locked, the waist went from (−3, 2, 6)° to (0, 0, 0) in about 0.1 s: torso pitch rate 1.0 rad/s, pelvis 0.78 rad/s, then 23 cm of backward stepping through the drag-in and lift, a 69 N left-palm spike, the left wrist forced open to 56° and the box tipping. The lock wrote the waist actuators' targets, but those actuators were not script-owned, so something else (most likely the motion graph) kept them at 0; the first recording had shown waist commands of 0 throughout the lock. While locked the waist actuators are now script-owned (`IK = false`) and given back at Release. After the fix the waist held its lock angles to Place, the jolt fell to 0.49 rad/s for 0.2 s, and a 0.2 m box reached Place and a 0.3 m box was released on the table with both palms on throughout the carry.

25.3 Squeeze changes (reverted)

- **Finger back-off.** When a palm came off its face while holding, the squeeze kept closing and loaded straight fingertips 16–22 cm from the 5 N·m wrist (40–50 N ≈ 8 N·m), forcing the wrists 20–30° open. Backing the hand off when the fingers alone exceeded *MaxFingerForce* was tried; an otherwise tick-identical 0.2 m run then loosened and slipped in the turn instead of reaching Place, so it was reverted.
- **First hand waits for the other.** Light boxes failed the squeeze because they slid 12–14 cm along the near-frictionless belt while the hands closed (3–5 cm in successes). A hand already touching the box held its gap, with its fingers still, until the other touched. It gave the first recorded success of a 0.45 kg 0.2 m box, but was reverted with the rest of the grasp changes (25.4).

25.4 The right hand, and the return to `82bc6f9`

The user saw the left arm jolt during the catch and sudden right-hand movements during the pre-grasp. The logs showed the parked right hand sagging about 12 cm below its target with its thumb pressing the conveyor's near edge at up to 222 N, then springing up at 0.85–1.5 m/s when the pre-grasp released it, which jolted the left hand holding the box. Folding the parked thumb, letting the parked wrist turn ±30°, parking 10 cm higher, a half-speed approach, and a hover over the belt beside the stop point were tried in turn. None fixed it cleanly.

At the user's request the whole grasp side was restored to commit `82bc6f9` (“every size box was working”): the catch, the move into and through the pre-grasp, the squeeze, wrist steering, finger and thumb poses and the parked hands' movement, which also drops the 28 October `ParkPose`, 5 cm/s parked hand and neutral parked wrist. Each of those functions was compared with `82bc6f9` and matches. The recording code, the DragIn and waist-lock fixes and everything after the grasp from `1515f8d` were kept.

Lesson

Each grasp-side change fixed one symptom on one box and moved the problem somewhere else; with episodes repeatable to the tick, a single run is not evidence. From here the grasp code stays at the `82bc6f9` baseline unless a specific change is asked for, and changes are judged on a jittered multi-box recording, not one run.

26. Determinism and why one run is not enough

- **Repeatable starts.** After the reset, episodes with the same box and start are identical to the tick (for example three `W40_L` episodes at 31.7 s and 1587 ticks each).
- **Hidden state still leaks.** Two `W40_L` episodes in different sessions started from identical first frames, yet one succeeded and one failed. The first difference was 0.0001 rad in the left elbow command at frame 18; it comes from the arm IK solver, which the reset does not clear, so it carries a trace of the previous episode. By frame 300 the joints differed by 0.26 rad. `W30_L` showed the same flip in reverse.
- **Light boxes are on a knife edge.** In light-box squeeze failures the box slid 12–14 cm along the belt during the grasp and started 1.5–2 cm further from the body; in successes it slid 3–5 cm.
- **Consequence.** Success is a rate per box, not a yes or no. Deal and jitter make each episode a different demonstration and give real rates.

27. Recording results, 30 October

Session	Setup	Success	Notes
16:10	1 episode, first recording	1/1	34.1 s, 1706 rows = sidecar ticks = video frames; ctrl one row later (3.5e-8 rad). Camera faced the wall.
16:18	3 episodes, `ResetToInitialState` only	1/2	Episode 1 started turned with the box at the robot's feet; stopped by hand.
16:23	qpos restored a frame early	1/3	Walker memory: identical starts, different failures.
16:46	Start-frame restore, fresh walker	3/3	Episodes 1 and 2 identical (38.1 s, placed 7.6 cm from the spot).
17:02–17:11	Pool, first versions	0	Stacked boxes exploded; wrong belt pose; check turned recording off; DragIn runaway found.
17:17	Pool, *BoxStart*/*ParkSpot*	2/3	`W30_L` slipped in FaceTable (left wrist twisted, palm off).
18:03	+ waist and drag fixes	4/6	`W30_L` failed the squeeze twice (identical); boxes repeated.
18:14	12 episodes, old build (no deal or jitter)	8/12	`W30_H` 4/4, `W40_H` 3/3, `W30_L` 1/1, `W40_L` 0/2, `W20_H` 0/1, `W20_L` 0/1.
20:22	**12 episodes, deal + jitter, grasp at `82bc6f9`**	**8/12**	No squeeze failures; QA 0 errors; all actions one row later.

The final session (20:22, commit `0700197`)

Box	Success	Failures
0.30 m, 0.67 kg (`W30_L`)	2/2	—
0.30 m, 1.48 kg (`W30_H`)	2/2	—
0.20 m, 0.45 kg (`W20_L` and `W40_L` at 0.2 m)	4/5	One slipped 5.5 cm at the start of the turn to the table.
0.40 m, 1.97 kg (`W40_H`)	1/2	One slipped 5.2 cm in Place.

Box	Success	Failures
0.20 m, 0.99 kg (`W20_H`)	0/2	Both slipped 5–6 cm at the start of the turn.

Mean episode 29.5 s: catch 7.3 s, grasp 4.0 s, lift 2.1 s, carry 12.0 s, place 2.5 s, retract 1.6 s. Starts moved 0.2–9.8 cm down the belt and up to 2 cm across. Viewer page: `~/Documents/LuckyEngine/Inspect/inspect_session_2026-10-30_20-22-04.html` .

28. Commits, state and known issues on 30 October

Commit	Date	Message	Contents
`82bc6f9`	27 Oct 18:42	Pipeline works for 0.1–0.4 m boxes; just placing is a bit slow	Section 19. The grasp baseline.
`1515f8d`	28 Oct 15:48	Good enough	Section 20; `data_pipeline/` and `plans/` (section 21).
`9de7a09`	30 Oct 16:51	Data collection works	Episode runner, MujocoFull reader, pelvis IMU, viewer (sections 22–23).
`986ee99`	30 Oct 18:09	Box pool randomization, drag-in and waist-lock fixes	Sections 24, 25.1, 25.2.
`0700197`	30 Oct 20:34	Data collection with box randomization works	Deal and jitter; grasp restored to `82bc6f9` (section 25.4).

Known issues.

- 0.2 m boxes slip out at the start of the carry turn (TurnToTable); the 1 kg one every time so far.
- A 2 kg 0.4 m box slipped once in Place.
- With the `82bc6f9` grasp, the parked right hand again sags onto the conveyor edge and springs up at the pre-grasp; the tall-box right-wrist wind-up (section 20.3) may return.
- The arm IK solver's state survives the reset, so identical starts can still diverge (section 26).
- *BoxStart* sits close to the belt's top end, so the jitter can only move the box down the belt.
- The scene's `Box_W40_L` is currently 0.2 m long.
- Not yet written: `G1ActAgent.cs` for closed-loop evaluation, and randomisation of the robot's start pose. `docs/ACT_explained.html` is not committed.
- The controller is 3,721 lines with 92 inspector fields, plus 305 lines and 9 fields in the episode runner. The engine copy still differs only in *ArmReach*.

Appendix A. Inspector fields added

Group	Field	Default	Meaning
Phases	CatchBox	true	Catch boxes sliding down the conveyor from where the robot stands (off: only boxes resting on the belt are picked).
Catch	CatchHandSpeed	0.5 m/s	Top speed of the stopper hand's IK target as it tracks its spot.
Catch	CatchShiftMax	0.15 m	How far the stop spot may move down the belt when the box would beat the hand.
Catch	CatchGive	0.06 m	Braking distance of the give once the box is on the palm.
Catch	CatchSpeed	0.03 m/s	Below this, a box against the palm counts as stopped.
Catch	CatchTime	0.3 s	How long it must stay stopped to count as caught.
Arms	StartShoulderDeg	(25, 10, 0)	Left shoulder pitch, roll and yaw at the start (right mirrored).
Arms	StartElbowDeg	−40	Elbow at the start (0 level, +90 straight, −60 folded).
Arms	HandLimit	93 %	Planning reach limit, measured from the standing pose.
Arms	IKTargetLimit	95 %	Per-frame IK target cap, measured from the live shoulder.
Arms	GripForceLimit	100 %	Share of the required squeeze each hand must hold for the grip to count.
Arms	GripComfort	85 %	Share of the arm the grip is planned within.
Arms	MaxSqueezeDepth	0.03 m	How far past its face a palm target may go.
Thumb	ThumbOpenDeg / ThumbGripDeg / ThumbTipDeg	41.25 / 0 / 0	Thumb open (not gripping), thumb in and thumb tip bend (gripping), clamped to the joint limits.
Tolerances	ReachTolerance	0.05 m	Grasp site this close to its target counts as in position.
Tolerances	ReachProgress	0.005 m	Smallest drop in hand error that counts as progress.
Tolerances	PlacedInside / PlacedFront / PlacedTop	0.015 / 0.03 / 0.01 m	The "hand is beside the box" checks.
Tolerances	PlacedWait	0.5 s	Hands beside the box must stop improving for this long before the grasp starts.
Tolerances	TopMargin	0.02 m	A raised grip keeps its sites this far under the box top.
Tolerances	AlignYawDeg / AlignSide / AlignForward	8° / 0.06 / 0.05 m	Approach alignment tolerances.
Tolerances	MinSide / MaxGripUp	0.11 / 0.15 m	Smallest grip-point distance from the centre line; highest grip point over the pelvis before a warning.
Tolerances	StillSpeed / StillTime	0.05 m/s / 0.5 s	The body counts as still below this speed, for this long.
Tolerances	TouchForce	2 N	A hand this loaded is at the box (also the palm-on threshold).

Group	Field	Default	Meaning
Tolerances	SlipTolerance / DropTolerance / LostDistance	0.01 / 0.05 / 0.25 m	Slip before boosting the squeeze, slip that counts as a drop, how far the box may move before it counts as gone.

Fields added 19–27 October

Group	Field	Default	Meaning
Catch	CatchPalmSide	0.17 m	How far out from the pelvis centre line the stopper palm stops the box, whatever its size.
Arms	HeldPalmForce	20 N	How hard each palm presses while holding (more if the weight needs it, never past MaxPalmForce).
Arms	MaxPalmForce	40 N	Most a palm presses: past it the squeeze twists the hand about the 5 N·m wrist yaw.
Arms	MaxFingerForce	25 N	Most the index and middle of a hand press; over it they ease back.
Arms	GripPoleOut	0.15 m	How far the elbow hint eases further out from the reach to the release.
Arms	LockWaist	true	From grip confirm to Place, the walker drives the legs only and the waist holds its angles.
Arms	LockWrists	false	Wrist servo off: the wrists hold where they are (a test switch).
Walk	CarryBackDistance / CarryBackSpeed	0.20 m / 0.4 m/s	The back-off from the conveyor with the box.
Walk	CarryTurnCmd / CarryTurnRamp	2.0 rad/s / 0.5 s	Walker yaw command for the carry turns, and its ease-in.
Walk	CarryWalkSpeed	0.7 m/s	Forward command walking the box to the table.
Walk	CarrySmoothing	0.2 s	Low-pass of the carry frame the box is held in.
Place	PlaceStandDistance	0.33 m	Pelvis to slot when placing.

Fields added, changed and removed 28–30 October

Group	Field	Default	Meaning
Arms	HeadClearance	0.25 m	A box tall enough to reach head height is carried with its near face this far in front of the head.
Arms	CarryOffset	(0.24, 0.10, 0)	Changed from (0.30, 0.10, 0): box centre while carrying, from the pelvis.
Arms	LockWaist	true	Now from grip confirm to Release (was to Place); the locked waist actuators are script-owned.
Place	PlaceEdgeInset	0.12 m	Now: the table spot the robot walks to is this far in from the near edge and ends.
Place	~~PlaceSpacing, MaxRepositions, RepositionStep, RepositionTolerance, MinStandDistance~~	—	Removed with the free-slot search and stepping closer.
Recording	RecordEpisodes	false	Record EpisodeCount episodes (needs a LuckyHub sign-in and the recorder set to MujocoFull with its cameras).
Recording	EpisodeCount	5	Episodes to record before stopping.
Recording	EpisodeTimeout	80 s	A run still going then ends the episode as failed (the recorder cuts at 90 s).

Group	Field	Default	Meaning
Recording	RandomizeBox	true	Deal the pool (every free box in the scene) like cards; off: box 0 every episode.
Recording	Seed	0	Seeds the deal and each episode's jitter (0 = from the clock, logged).
Recording	BoxStart	(0.616, 0.734, 0.978)	World centre of the episode's box at the start (top of the belt).
Recording	StartJitter	(0.10, 0.02) m	Up to X down the belt (never up) and ±Y across it.
Recording	ParkSpot	(−2, 0.1, −1.5)	World spot of the first box not in use; the rest line up 0.6 m apart along +Z.
Recording	SidecarDirectory	""	Where run_<time> sidecar folders go ("" = ~/Documents/LuckyEngine/DataSidecars).

The scene's saved values differed in places, for example MinSide 0.08, MaxSqueezeDepth 0.05, GripComfort 95, ThumbOpenDeg 30, ApproachGap 0.02, BeltClearance 0.02, MaxBoxAttempts 5, WalkToBox false.

Appendix B. Key formulas

Quantity	Formula
Required squeeze per hand	GripSafetyFactor · m · g / (2·BoxFriction) · boost
Box acceleration on the belt	a = g(sinθ − μcosθ); frictionless arrival speed v = √(2 g sinθ d)
Distance a knocked box slides sideways	v^2 / (2μg)
Downhill direction in the belt plane	d = normalize(n · n_y − up), |n·n_y − up| = sinθ
Box arrival time at the stop spot	t = (−v + √(v^2 − 2a·past)) / a, with past < 0 while the box is still uphill
Speed-matched give (approach)	v_{palm} = v_{box}(1 − √(min(1, gap / 0.05 m))): meets the box at its speed after about 2×0.05/v seconds and 5 cm of palm travel
Give braking	deceleration = v_{palm}^2 / (2 · CatchGive)
Stopper tracking step	move = min(dist · (1 − $e^{-dt/0.1\ s}$), CatchHandSpeed · dt); in the give, move = dist
Holder offset (HoldHere)	d = Cmd − HandTarget; offset = (d·lat, d·fwd, d_y); target = HandTarget + offset in the current grip frame
Squeeze while holding (per palm)	max(req, min(HeldPalmForce, MaxPalmForce) / 1.25), with req = min(GripSafetyFactor·m·g/(2·BoxFriction)·boost, MaxPalmForce/1.25); the gap aims the palm at 1.25× that
Twist on a wrist from the squeeze	τ ≈ F_{palm}·0.08–0.11 m + $F_{fingers}$·0.16–0.22 m, against 5 N·m (pitch and yaw)
Carry stop distances	back-off stops at max(d − v·0.4 s, d/2); walk stops 0.10 m · (v / 0.6)² short
Station hold (standing with the box)	start when |pelvis − anchor| > 0.10 m, still for StillTime and not turning; one axis: forward/back if |fwd error| > 0.04 m, else sideways; stop below 0.04 m, on overshoot or when a turn starts
Pelvis IMU from the recorded pose	gravity = R^T(0, 0, −1); ω_k = 2·fps·sign(q_{k-1}·q_k)·vec(q_{k-1}^* q_k), 0 on the first row
Recorded action alignment	action(state row k) = ctrl of row k+1; walker command and waist lock from sidecar tick k
Pool deal	round r = ⌊e / n⌋; order = Fisher–Yates shuffle with seed 7919·BaseSeed + r; box = order[e mod n]
Start jitter	start = BoxStart + down·u·J_x + (n × down)·(2v − 1)·J_y, u, v ~ U(0,1) from seed BaseSeed + e; down = normalize(n·n_y − up)
Box placed in MuJoCo	p_{Mj} = A·(p_{world} − sceneTranslation), q_{Mj} = A for an upright box, A = GetAxisHzToMj() = (0.707, 0.707, 0, 0)

Quantity	Formula
Twist resisted by torsional friction	$\tau = \mu_{torsion} \cdot N$ per contact, against the box's weight × its offset from the grip line